MATH 128a: Homework #10

Instructor: F. Weber

Due by: **Wednesday, November 12, 2025, 11:59pm**

You may use any Matlab functions you find on the bcourses website to solve these problems. If you do, make sure to mention it.

Problem 1: Problems from the textbook, chapter 3.4

Solve problems 2a, 4a, 11 of Section 3.4 of the textbook.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 2: Problems from the textbook, chapter 4.1

Solve problems 2a, 4a, 5d, 24(22) of Section 4.1 of the textbook.

In 24(22), use a linear combination of $f(x_0 - h)$, $f(x_0)$, $f(x_0 + h)$, $f(x_0 + 2h)$ and $f(x_0 + 3h)$ (i.e., also use $f(x_0)$ contrary to what the hint suggests)
(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 3: Problems from the textbook, chapter 4.2

Solve problems 1a, 5, 9 10 of Section 4.2 of the textbook.

In 5: use the notation of Lecture 19, i.e. assume $N_2(h/2^j)$ has the given values, that $M = N_2(h) + K_2h^2 + K_4h^4 + K_6h^6 + K_8h^8 + O(h^{10})$, and determine $N_8(h)$.

(If the problem has a different number in the 9th edition book, it is in parentheses)

Problem 4: Derivation of finite difference formula

Let $f : \mathbb{R} \to \mathbb{R}$ be a smooth, bounded function and $\bar{x} \in \mathbb{R}$. A one-sided difference approximation to $f'(\bar{x})$ based on $f(\bar{x})$, $f(\bar{x} - h)$ and $f(\bar{x} - 2h)$ is given by

$$D_2 f(\bar{x}) = \frac{3f(\bar{x}) - 4f(\bar{x} - h) + f(\bar{x} - 2h)}{2h}. \tag{1}$$

Determine the quadratic polynomial $p(x)$ that interpolates f at the points $\bar{x}$, $\bar{x} - h$ and $\bar{x} - 2h$. Verify that evaluation $p'(\bar{x})$ gives equation (1).

Problem 5: Use of `fdstencil`

a) Use the method from Lecture 19 with the Taylor expansion to set up the 5×5 linear system that would determine a fourth-order accurate finite difference approximation to $f''(x)$ based on 5 equally spaced points,

$$f''(x) = c_{-2} f(x - 2h) + c_{-1} f(x - h) + c_0 f(x) + c_1 f(x + h) + c_2 f(x + 2h) + O(h^4).$$

b) Compute the coefficients using the MATLAB code `fdstencil.m` available from the website, and check that they satisfy the system you determined in part (a).

c) Test this finite difference formula to approximate $f''(1)$ for $f(x) = \sin(2x)$ with values of h from the array `hvals = logspace(-1, -4, 13)`. Make a table of the error vs. h for several values of h and compare against the predicted error from the leading term of the expression printed by `fdstencil`. guidance on how to make such a table.

Also produce a log-log plot of the absolute value of the error vs. h.

You should observe the predicted accuracy for larger values of h. For smaller values, numerical cancellation in computing the linear combination of u values impacts the accuracy observed.

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